

MeshAI

Run big models on the devices you already own.

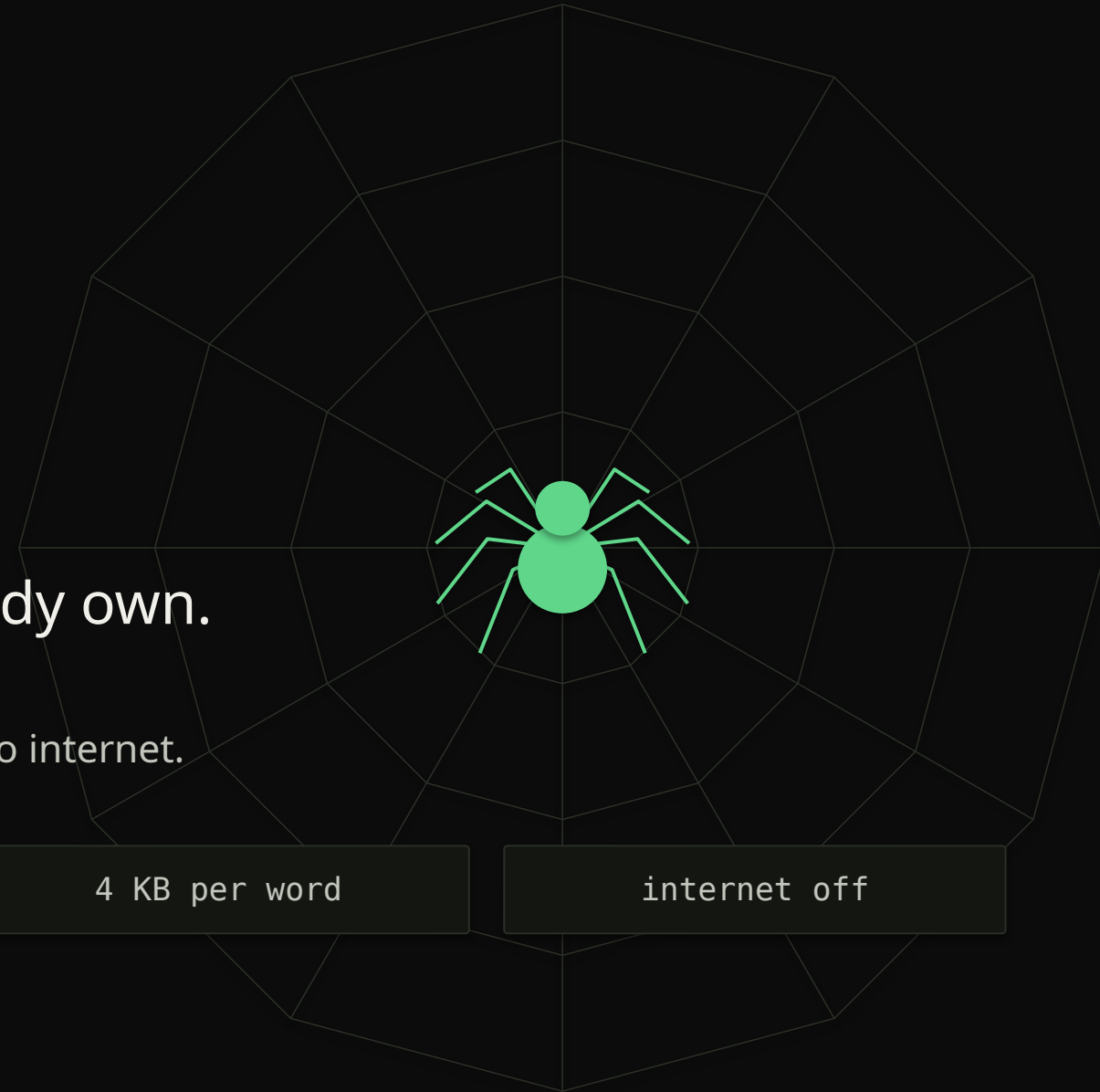
A 30B model across one laptop and two phones. No cloud. No internet.

ready in 70 s

6.6 tokens per second

4 KB per word

internet off



A 30B model needs 18.6 GB of memory.

If it does not fit, you cannot run it. So people run small models, and small models get it wrong.

WHAT A MODEL NEEDS

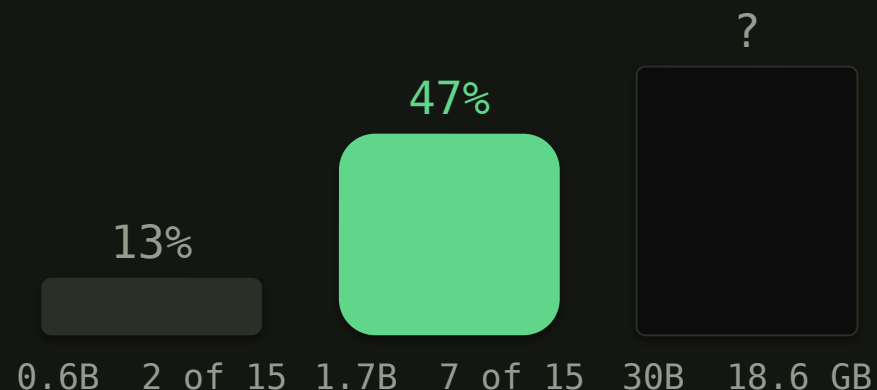
Qwen3 0.6B	0.64 GB
Qwen3 8B	4.7 GB
gpt-oss 20B	12.1 GB
Qwen3-Coder 30B	18.6 GB

WHAT A DEVICE HAS FREE

A budget laptop	about 4 GB
One 16 GB phone	about 8.7 GB
The three together	about 20 GB

SMALL MODELS GET IT WRONG

We asked three setups the same 15 coding problems and ran their answers against the tests.



**So how do you run a model
that does not fit?**



Phones and a laptop, as one machine.

One device plans and answers. The others hold part of the model. Your tools do not change.

- Phones and laptops in one cluster
- One device decides who holds what
- Only about 4 KB crosses the cable per word
- It says no when a device would slow you down

THE PHONE IS NOT AN ACCESSORY

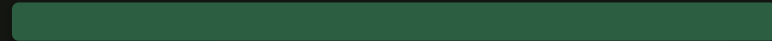
A phone can be the host: it plans the split, runs the model and answers.

Every other system treats a phone as a client that asks a server.

Layers split across the devices.

THIS LAPTOP · THE HOST

Embeddings and output head stay here



Layers 0–22 · 9.19 GB

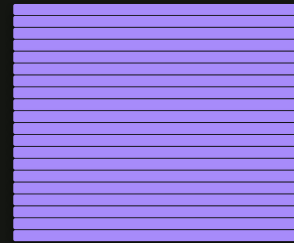


Plans the split · serves one endpoint · reads the model file once

A PHONE

Layers 23–42

7.53 GB

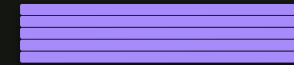


keeps its layers

A PHONE

Layers 43–47

2.04 GB



ready in 70 s

48 layers

one pass, in order

4 KB

crosses each cable, per word

18.6 GB

of weights never move

2–3 ms

over the USB cable

The host decides

from live memory, heat, battery and link speed

The phone never sees your prompt

it gets numbers, computes, sends numbers back

The head stays home

which is why only kilobytes cross

Loading went from 517 s to 70 s.

Loading the model

517 s → 70 s

It was reading the file in scattered pieces. Now it reads it once, start to finish.

A split it has not used

437 s → 70 s

Each device now keeps its own layers, checked against the original file.

First word in a long chat

slow → fast

One conversation slot, so the chat stays in cache instead of being read again.

AND THE REST

- The head stays on the host
- We tested it hot, not cold
- We take the middle ping, not the average
- Photos and text on two phones at once

Use it when the model does not fit.

It is worth knowing where this helps, because it is the same thing as knowing where it does not.

USE IT WHEN

- The model will not fit on one device
- The work cannot leave your devices
- You have devices sitting idle
- You run it often

DO NOT USE IT WHEN

- The model already fits one device
splitting would only make it slower, and the app
- ^{refuses} You want the fastest possible answer
every word crosses the cable once, so a cloud is
- ^{quicker} Your data is free to leave
then a cloud API is cheaper per word, and we say so
- You need a stranger's phone to be private
we only claim privacy across devices one owner controls

Review your code before you commit.

One tap on the phone. The answer comes from a 30B model running on your own devices.



WHY PEOPLE KEEP IT

- It installs as a git hook, in one command
- Every commit gets read before it lands
- No bill, however many times you run it

AND IT NEVER LEAVES

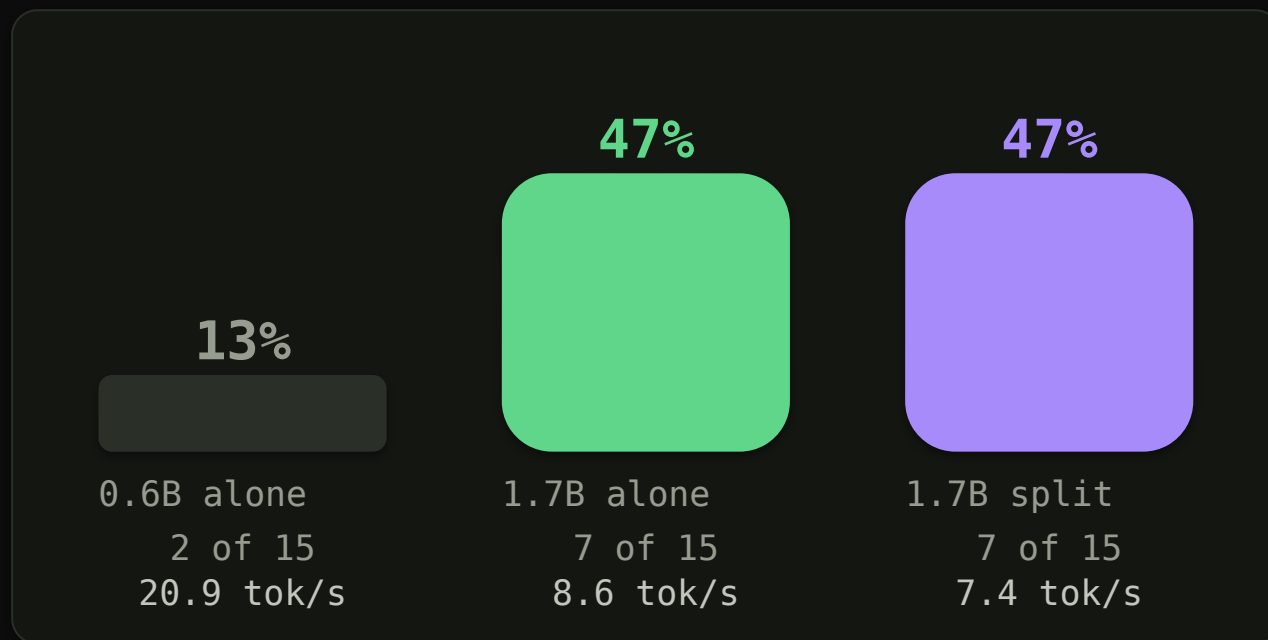
- Internet off. No account. No upload
- The code stays on the desk it was written on
- Also: review a file, or write its tests

**Does splitting it
make the answers worse?**



13% to 47%, on the same tests.

Same 15 coding problems for every setup. We ran each answer against its own tests. Right or wrong, nothing in between.



WHAT THIS SHOWS

- A bigger model is three times better
- Splitting costs a little speed
- Splitting does not change the answers

Speed drops a little. Accuracy jumps a lot.

Nobody else pools phone memory.

SYSTEM

Petals

arXiv 2209.01188

prima.cpp

arXiv 2504.08791

EdgeShard

arXiv 2405.14371

exo

github, 47k stars

Darkbloom

Eigen Labs

llama.cpp RPC

we build on it

Ollama, LM Studio

and the rest

WHAT IT DOES

A public swarm of volunteer servers

Splits a model across a home cluster

Splits across edge devices

Splits across your Macs and Linux boxes

Rents idle Macs and pays the owners

Runs layers on another machine

Great on one machine, one model

WHY IT IS NOT THIS

0.83 words a second, across continents

Leaves phones out on purpose

Jetsons and servers, not phones

A phone is not a real device in it

One whole model per Mac. Never a phone

You type the split by hand

If it does not fit, you cannot run it

MeshAI

Phones and laptops together. It decides who holds what, and says no when it should.

MeshAI

a search of arXiv for multi-phone inference returns nothing

**Who pays for this,
and what are they buying?**



We rent memory.

The models are open and free. What people do not have is the memory to hold them. A phone in a drawer is cheap memory.

THE FLEET

People give us a phone they no longer use. We keep it, and pay them.

- Racked and plugged in, in our hands
- No battery to ruin, no pocket to leave
- We buy no hardware. They earn from a drawer

100 old phones is about 800 GB of memory.

WHAT A CUSTOMER BUYS

Enough memory to hold the model they want to run.

A 8B model	about 6 GB
A 20B model	about 13 GB
A 30B model	about 20 GB
A 70B model	about 45 GB

Bigger model, more memory, higher tier.

AND TWO MORE WAYS

Their own devices

a lab or a campus turns its idle phones into a private cloud. Sold per site.

Licensed to a phone maker

two phones are worth more together than apart, which is a reason to own the second.

That last buyer is in this room.

It ran this weekend.

WHAT RAN

Qwen3-Coder 30B

- 18.6 GB across a laptop and two phones
- The phones held 9.6 GB of it
- Ready in 70 seconds
- 6.6 words a second
- Three coding questions, all correct
- Internet off the whole time

BUILT IN THE WINDOW

- llama.cpp built for phones
- The app: host and helper
- The laptop agent and its panel
- Our own model reader and planner
- The layer store
- The test harness
- The CLI and the git hook

IF SOMETHING GOES WRONG

- | | |
|-----------------------|--------------------------------|
| Slow link | refused above 60 ms |
| Phone short of memory | we reserve, and you can cap it |
| A device drops | |
| Engine missing a flag | we ask it what it has |
| Bad model file | every read is checked |
| A strange machine | one button checks everything |

It says no before it disappoints you.

Thank you

A big model, on the devices you already own,
with the internet switched off.

Prajwal Gunnala · Yuva Raj Ambati

github.com/prajwal-gunnala/maynards

